

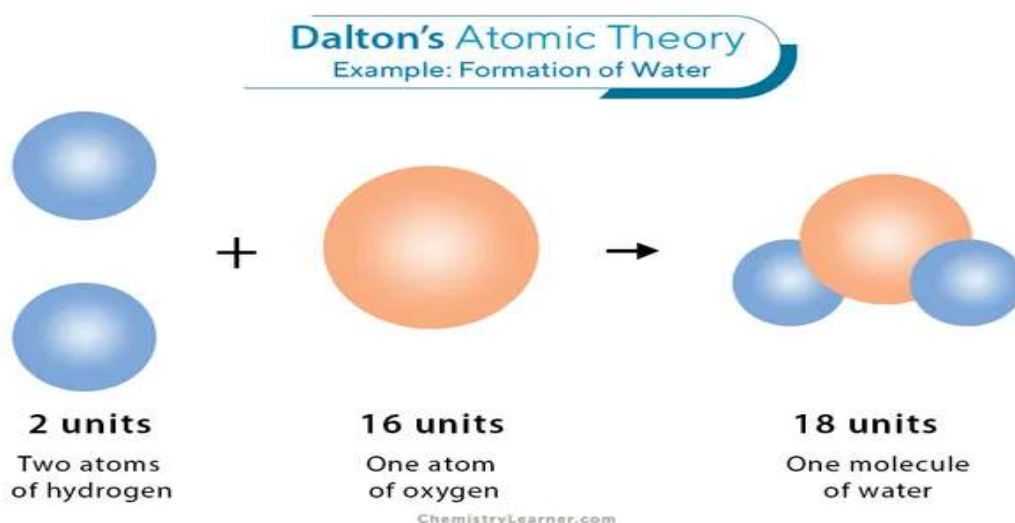
Basic Of Atomic Theories

Introduction:

The development of atomic theory progressed from Democritus' ancient idea of indivisible particles to John Dalton's foundational chemical model. J.J. Thomson's discovery of the electron led to the Thomson' model which was overturned by Ernest Rutherford's nuclear model. Niels Bohr introduced quantized electron orbits, culminating in the modern quantum mechanical model, which describes electrons as existing in probabilistic clouds around a dense nucleus.

Atomic Models:

Dalton's Atomic Theory (1803):

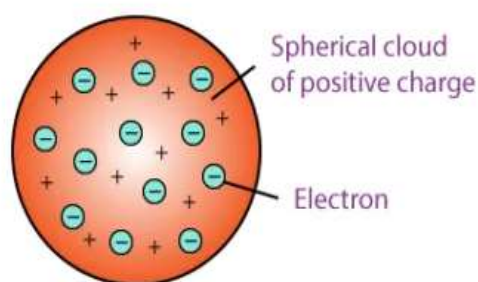


Postulates of Dalton's Atomic Theory

1. All matter is made up of extremely small particles called atoms.
2. Atoms are indivisible and indestructible.
3. All atoms of a given element are identical in mass and properties.
4. Atoms of different elements differ in mass and properties.
5. Compounds are formed when atoms of different elements combine in simple whole-number ratios.
6. Compounds are formed when atoms of different elements combine in simple whole-number ratios.

Limitations of Dalton's Atomic Theory

1. Atoms are divisible and contain subatomic particles.
2. Could not explain isotopes.
3. Failed to explain electrical nature of matter.
4. Did not describe atomic structure.

Thomson's Atomic Model (1897) :

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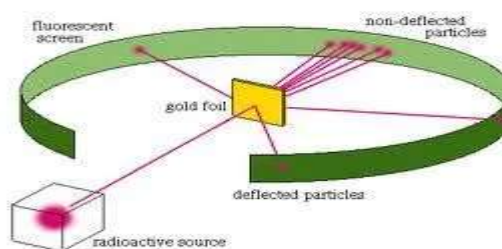
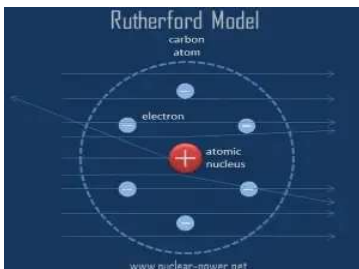
Postulates of Thomson's Atomic Model

1. The atom is a uniform sphere of positive charge
2. Electrons are embedded in the positive sphere.
3. Mass is uniformly distributed.
4. Electrons are embedded in the positive sphere.

Limitations of Thomson's Atomic Model

1. Failed to explain Rutherford's gold foil experiment.
2. Could not explain concentration of positive charge.
3. Did not explain atomic stability.
4. Failed to explain atomic spectra.

Rutherford's Atomic Model (1911) :



Definition

Rutherford's atomic model states that an atom contains a small, dense, positively charged nucleus at its center, with electrons revolving around it.

Postulates of Rutherford's Atomic Model

1. Atom has a tiny, dense, positively charged nucleus.
2. Nearly all atomic mass is concentrated in the nucleus.
3. Electrons revolve around the nucleus in circular paths.
4. Most of the atom is empty space.
5. Electrostatic attraction holds electrons around the nucleus.

Limitations of Rutherford's Atomic Model

1. Could not explain stability of atoms.
2. Failed to explain line spectra of elements.
3. Contradicted classical electromagnetic theory.
4. Did not explain electron arrangement.

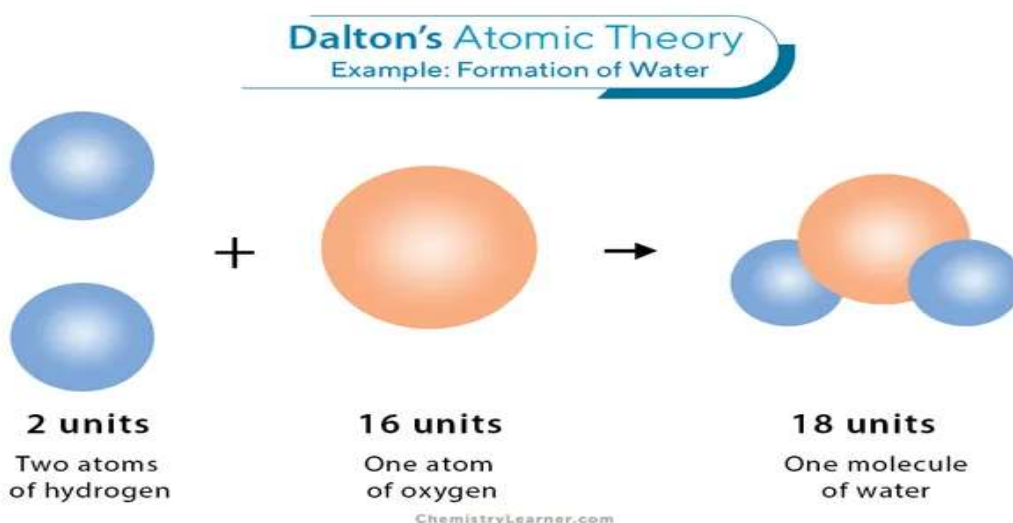
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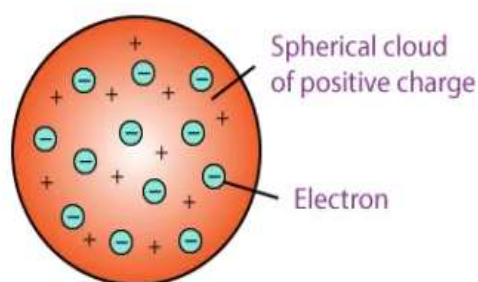


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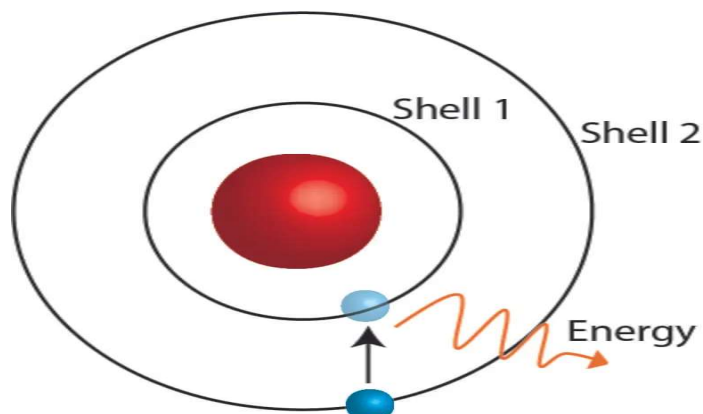
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Bohr's Atomic Model (1913) :



Definition

Bohr's atomic model states that electrons revolve around the nucleus in fixed circular orbits of definite energy without radiating energy.

Postulates of Bohr's Atomic Model

1. Electrons move around the nucleus only in certain permitted circular orbits called stationary states.
2. Each orbit has a fixed and definite energy.
3. Electrons do not lose or gain energy while moving in a stationary orbit.
4. Energy is absorbed or emitted when an electron jumps from one orbit to another.
5. The lowest energy orbit is called the ground state.

MATHEMATICALY

Quantization of angular momentum:

$$mvr = \frac{2\pi nh}{2\pi}$$

Energy change during transition:

$$\Delta E = E_2 - E_1 = h\nu$$

Limitations of Bohr's Atomic Model

1. Applicable only to hydrogen and hydrogen-like atoms.
2. Cannot explain the fine structure of spectral lines.
3. Does not explain Zeeman and Stark effects.
4. Violates Heisenberg's uncertainty principle.

5. Does not explain the wave nature of electrons.
6. Cannot explain chemical bonding and molecular structure.

Introduction To Particle Physics

Introduction:

Particle physics is a branch of physics that studies the fundamental particles of matter and the forces through which they interact. It seeks to understand the basic building blocks of the universe, such as quarks, leptons, and gauge bosons, and explains how these particles combine to form atoms and larger structures. Particle physics also explores the fundamental forces—electromagnetic, weak, strong, and gravitational—that govern the behavior of matter at the smallest scales.

History:

The history of particle physics began in the late 19th and early 20th centuries with the discovery of the electron, proton, and neutron. The development of quantum mechanics and relativity laid the foundation for understanding subatomic phenomena. Later discoveries of mesons, neutrinos, and quarks led to the formulation of the Standard Model in the 20th century, which successfully explains the behavior of fundamental particles and forces. Modern particle physics continues to advance through high-energy experiments using particle accelerators and detectors.

Standard Model

Definition:

The Standard Model is a well-established theoretical framework that describes the fundamental particles of nature and the interactions between them, excluding gravity. It classifies all known elementary particles into **fermions** (quarks and leptons), which make up matter, and **bosons**, which act as force carriers. The model explains three fundamental forces: **electromagnetic**, **weak**, and **strong** interactions, mediated by the photon, W and Z bosons, and gluons respectively. The **Higgs boson**, discovered in 2012, explains how particles acquire mass. The Standard Model has been highly successful in predicting experimental results, although it does not explain gravity, dark matter, or dark energy.